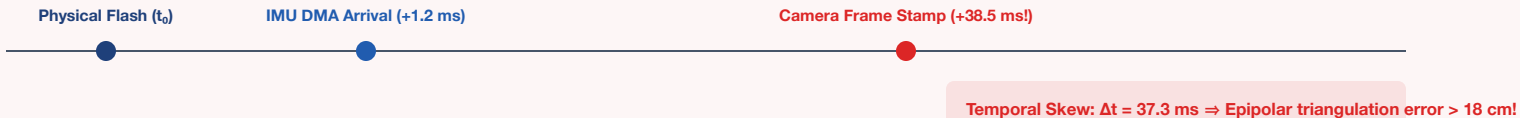


SENSOR SYNCHRONIZATION: HARDWARE PTP VS SOFTWARE STAMP ILLUSION

Why Linux User-Space Timestamping Corrupts Multi-Modal Fusion and Injects Spatial Drift

× THE SOFTWARE TIMESTAMP ILLUSION (`gettimeofday` / ROS Header)

Timestamps assigned after OS scheduling, driver context switches, and USB/Ethernet queue delays



✓ HARDWARE PTP / IEEE 1588 SYNCHRONIZATION (Cross-Triggered)

Timestamps latched in silicon hardware registers directly at photonic / IMU sample strobe

